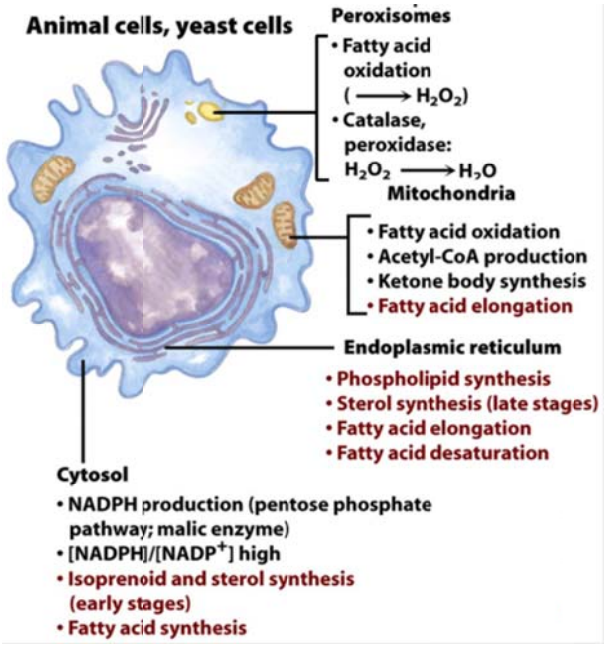
# BIOSYNTHESIS OF FATTY ACIDS AND TRIACYLGLYCEROLS

- Lipids are involved in many cellular functions and make the main source of energy for most living beings
- Among its multiple functions are:
    - pigments (retinal, carotenes)
    - cofactors (vitamin K)
    - detergents (bile salts)
    - hormones (derivatives of vitamin D, steroids)
    - cellular messengers (inositols, eicosanoids)
    - protein anchoring to membranes (myristoyls, palmitoyls)



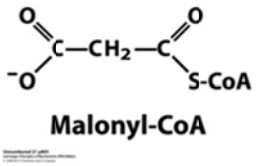

- From all this we can conclude that the capability of cells to synthesize lipids is essential
- In general, the reactions used in lipid synthesis processes are endergonic (require energy) and consume reducing equivalents (mostly NADPH)
- Both the synthesis and the oxidation of lipids are compartmentalized in different regions of the cell
- **The synthesis of fatty acids** and some estages of the synthesis of **cholesterol occurs in the cytoplasm**.
- Furthermore, the oxidation of fatty acids occurs in the mitochondria and, in the case of long fatty acids, partially in the peroxisomes

## Fatty Acid Biosynthesis

- The biosynthesis and degradation of fatty acids are independent processes, catalyzed by a different set of enzymes and takes place in different cell compartments.
-In humans, excess glucose and carbohydrates are transformed into pyruvate and Ac.CoA
-Precisely the Ac. CoA is the necessary precursor for the synthesis of fatty acids.
-Later these fatty acids are stored in the adipose tissue.
-The tissues that have a more active synthesis of fats are the adipose, liver and mammary glands (lactation)

-The synthesis takes place in the cytoplasm and is catalyzed by an enzymatic complex with six enzymatic activities called **Fatty acid synthase**.

- The precursor molecule of all the carbons of the fatty acid is **malonyl CoA.**



Malonyl-CoA

- This compound comes from the carboxylation of Acetyl CoA, so in net terms it can be said that all the carbons of an ac. fatty **have their origin in Ac.CoA**
- This is the moment to remember that **malonyl-CoA is precisely the regulator of the β-oxidation of fatty acids through the allosteric inhibition of CATI (carnitine acyl transferase I)**.

**Source of carbons and reducing power**

- The main source of carbons for fatty acids biosynthesis comes from Ac. CoA generated from the oxidation of glucose to pyruvate and also from the catabolism of the carbon skeletons of amino acids.
- In general, the message is that the surplus of glucose is transformed in fatty acids and therefore in triacylglycerols.
- It is insulin that will facilitate the activation of enzymes that promote the synthesis of fatty acids and lipids in general.
-Ac. CoA produced during the oxidation of fatty acids does not represent a significant source of Ac.CoA for the biosynthesis of fatty acids in animals, since both routes are reciprocally regulated.
- Since the synthesis of fatty acids takes place in the cytoplasm and Ac. CoA is produced in the mitochondria, is necessary a shuttle system to allow the transport of carbons from the mitochondrial matrix to the cytoplasm.
- For this the. intramitochondrial Ac CoA reacts first with OA to form citrate (citrate synthase).
- The citrate then pass out the mitochondria to the cytoplasm through the citrate transporter.
- Once in the cytoplasm, citrate lyase regenerates Ac. CoA and OA.

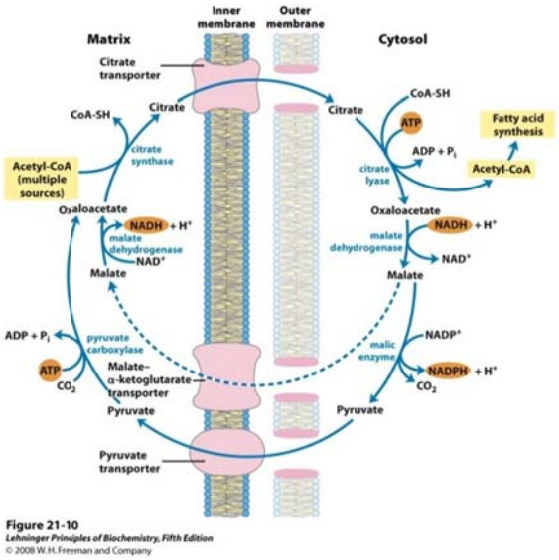


Figure 21-10
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- The OA cannot return to the mitochondria directly since there is not a transporter for OA; In order to do this, malate DH reduces OA to malate, which can be returned to the mitochondria via the malate-α-ketoglutarate transporter,.
- In the matrix the malate is reoxidized to OA to complete the exchange.

- However, in order to achieve some of the reducing power ,in the form of NADPH, necessary for the subsequent synthesis of fatty acids, cytoplasmic malate is decarboxylated and oxidized to pyruvate by the malic enzyme.
- In this case, pyruvate is transported to the mitochondria by the pyruvate transporter and is converted back to OA by pyruvate carboxylase (which in this situation will be activated by the presence of high amounts of Ac. CoA).
- In this transport system, two ATPs (citrate lyase and pyruvate carboxylase) are consumed for each molecule of Ac CoA submitted for the synthesis of fatty acids.
-With the malic enzyme,the halfpart of NADPH necessities are covered and the rest comes from the pentose phosphate pathway.

**Synthesis of Malonyl CoA**

- The formation of malonyl-CoA from Ac. CoA is an irreversible process catalyzed by **acetyl-CoA carboxylase (ACC)**.
- This enzyme has three activities in a single multifunctional polypeptide.
- The enzyme has a biotin prosthetic group bound to the ε-amino group of a Lys.
- The reaction is very similar to other carboxylations such as that catalyzed by pyruvate carboxylase

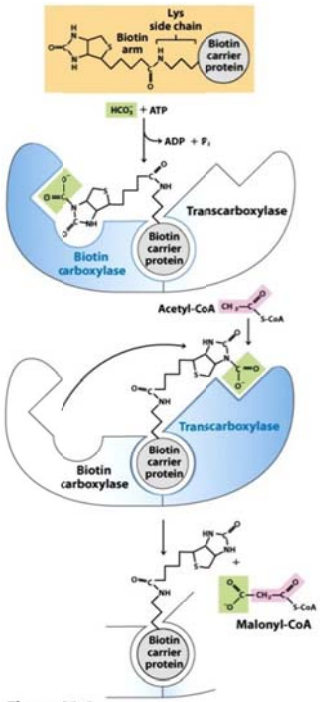



**Figure 21-1**
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**Regulation of the acetyl-CoA carboxylase**

- When a cell has more fuel than required for its needs, the excess generally in converted in fatty acids and stored as TAG
- The ACC reaction is the limiting step of fatty acid biosynthesis rate.
- The palmityl CoA, main product of the synthesis of Fatty acids is an inhibitor while citrate is an activator.
- When there is an increase in Ac.CoA and ATP in the mitochondria, citrate is transported to the cytoplasm and becomes cytoplasmic AcCoA that generates an allosteric activator signal to ACC.
- At the same time, citrate inhibits PFK1 thereby reducing carbon flux via glycolysis.
- ACC is also regulated by covalent modification.

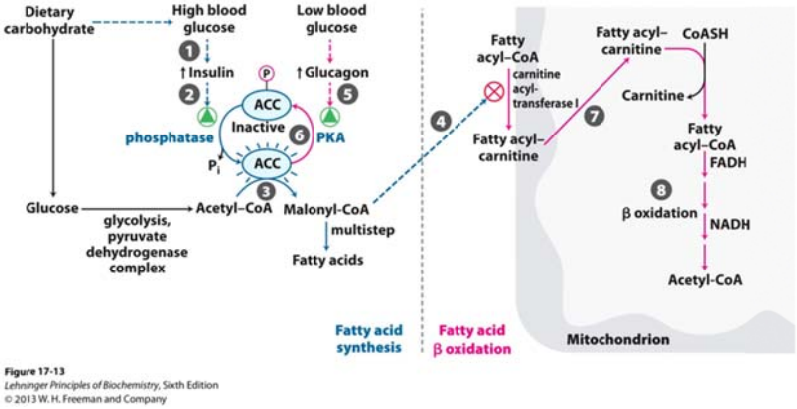



Figure 17-13
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- The phosphorylation of ACC induced by glucagon (hypoglycemia) and epinephrine, inactivates the enzyme which makes the synthesis of fatty acids slower but, at the same time, favors the transport of fatty acids to the mitochondria for oxidation .
- In contrast, insulin (hyperglycemia) promotes ACC dephosphorylation; in this dephosphorylated stat, ACC is active thereby generating malonyl-CoA and stimulating fatty acid synthesis

**Fatty Acid synthesis mechanism**

- The mammalian fatty acid synthase (FAS I) is made up of a single multifunctional polypeptide chain
- Contains six active sites for six different reactions located in separate domains
- There is a seventh domain without a catalytic function but contains a prosthetic group attached to it that serves as an anchoring site for the growing fatty acid chain throughout the process. It is called Acyl Carrier Protein (ACP)

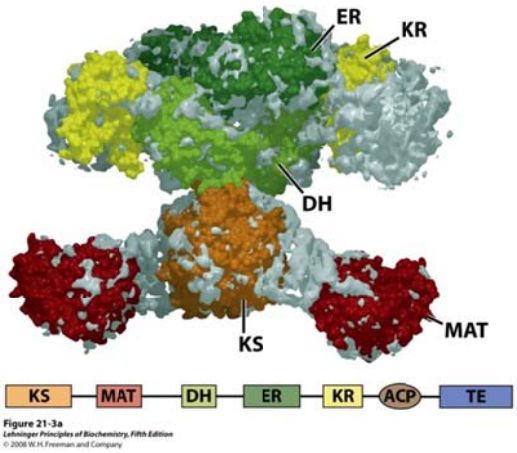


Figure 21-3a
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- This prosthetic group linked to ACP is the phosphopantetheine

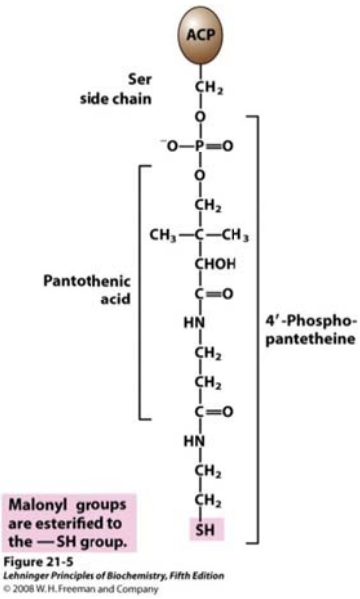


Figure 21-5
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- The enzymatic activities present in ac. fatty synthase are:
    - -KS: ketoacyl ACP synthase
    - MAT: malonyl / acetyl-CoA transferase
    - DH: hydroxyacyl ACP dehydratase
    - ER: enoyl-ACP reductase
    - KR: ketoacyl-ACP reductase
    - TE: thioesterase
- The synthesis of fatty acid occurs in the form of cycles wiht four reactions that are repeated until the fatty acid has 16C (palmitic)
- The reactions are:
    - Condensation
    - Reduction
    - Dehydration
    - Reduction
- Before the reactions of the first cycle can begin, the two thiol groups present in the ACP and in the KS (synthase priming) must be loaded with AcCoA and MalonylCoA.
- In the first place, the acetyl group of the Ac.CoA is transferred to the ACP in a reaction catalyzed by MAT
- This acetyl group is then transferred to the thiol of the KS
- Secondly, the malonyl group is transferred from malonyl-CoA to the thiol of ACP.
- In this way the enzyme is uploaded and ready for the condensation reactions of both groups

1) **Condensation:** the free carboxyl group of the malonyl bound to the ACP is decarboxylated leaving a methylene radical that reacts with the acetate bound to the KS. In this way a ketobutyryl-ACP is originated. This reaction is catalyzed by KS

Note that the $CO_2$ released in this reaction is the same that had been introduced in the Ac.CoA activation reaction catalyzed by ACC

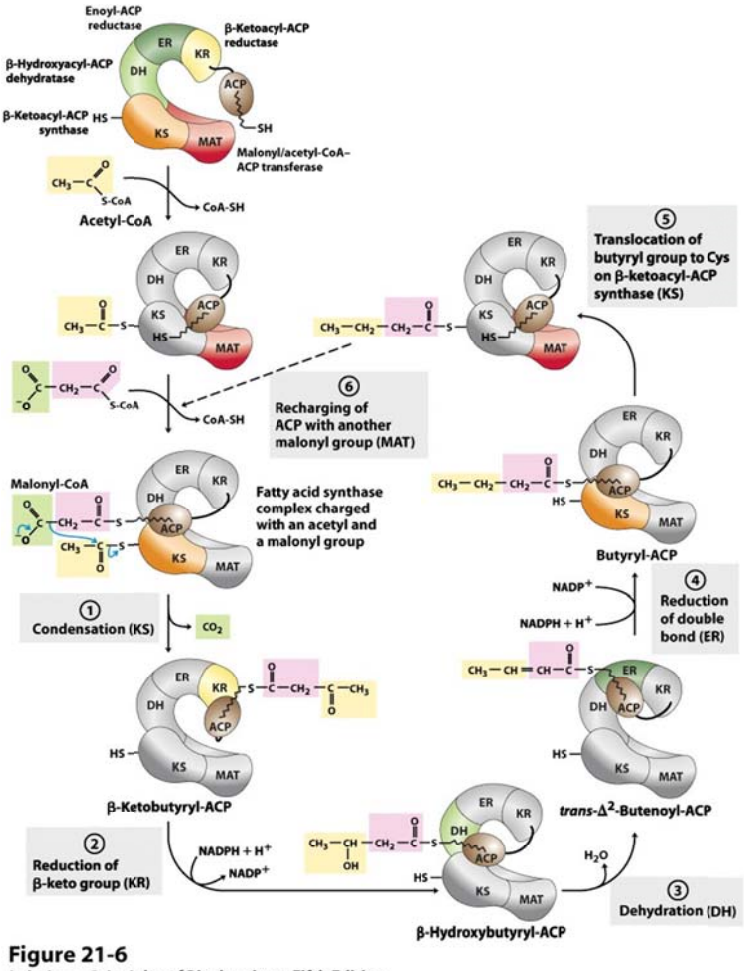



**Figure 21-6**
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2) **Reduction**: the carbonyl group of C3 is reduced by NADPH and generates a β-hydroxybutyryl-ACP. This reaction is catalyzed by KR.

3) **Dehydration**: a molecule of $H_2O$ is eliminated from the elements of C2 and C3, which generates a trans$\Delta^2$-butenoyl-ACP. This reaction is catalyzed by DH

4) **Reduction**: finally the double bond is reduced through the use of NADPH, which generates a four-carbon fatty acid linked to ACP, butyryl-ACP. This reaction is catalyzed by ER.

- For a new cycle to start, it is necessary to transfer the butyryl from the ACP to the KS. This is done by KS through a transesterification reaction.
- This movement of the fatty acid frees the ACP so that it can receive a molecule of malonylCoA
- From this moment on, a condensation can occur between the butyryl group and the two carbons of the malonyl-ACP group with the concomitant release of $CO_2$, and the rest of the reactions of the synthesis cycle.
- Once the fatty acid linked to the ACP has 16C, the activation of the thioesterase activity (TE) occurs, thereby releasing the fatty acid from the enzyme.
- Therefore, the majority end product of the FASI is Palmitate (16C)
- A special situation in the synthesis of fatty acids occurs in women during pregnancy and lactation. Maternal milk contains short chain fatty

acids (less than 16C). In this situation, as a consequence of the hormonal profile of the woman, TE activity is earlier so the fatty acids synthesized are shorter.

**Overall reaction for the synthesis of palmitate**

1) Production of 7 molecules of malonyl-CoA
   7 Ac-CoAs are carboxylated to 7 malonyl-CoA.

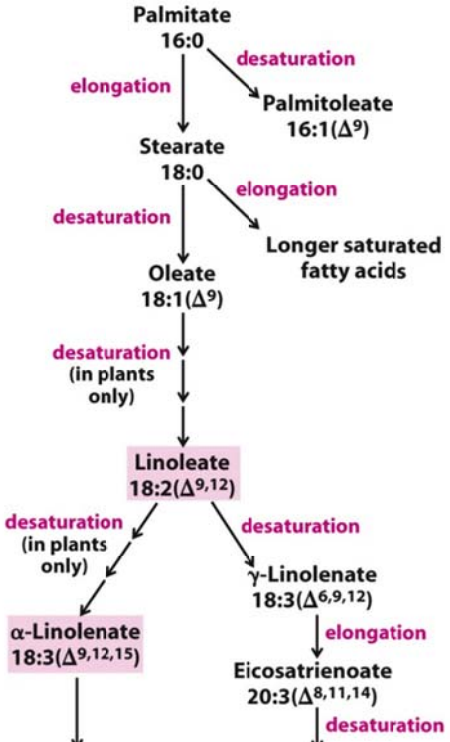
$$7\ AcCoA + 7\ CO_2 + 7\ ATP \rightarrow 7\ Malonyl\text{-}CoA + 7\ ADP + 7\ P_i$$

2) Seven cycles of FASI: condensation, reduction, dehydration and reduction using NADPH

$$AcCoA + 7\ Malonyl\text{-}CoA + 14\ NADPH + 14\ H^+ \rightarrow Palmitate\ (16C) + 7\ CO_2 + 8\ CoA + 14\ NADP^+ + 6\ H_2O$$

3) Additon of 1) and 2) give us the final balance

$$\mathbf{8\ Ac\text{-}CoA + 7\ ATP + 14\ NADPH + 14\ H^+ \rightarrow Palmitate + 8\ CoA + 14\ NADP^+ + 6\ H_2O}$$



## Long-Chain Fatty Acids

- Palmitate is the major end product of FASI in animal cells, and it is also the precursor of other longer fatty acids.
- The elongation of palmitate occurs through successive additions of acetyl groups by the action of enzymatic elongation complexes, called elongases, which are located in the REL and in the mitochondria
- The chemical mechanism for the elongation process is identical to that of fatty acid synthase.

## Unsaturated fatty acids

- Palmitate and stearate are the precursors of the two most abundant monounsaturated fatty acids in animal tissues: palmitoleate (16: $1\Delta^{9}$) and oleate ($18.11\Delta^{9}$)
- The cis double bond is introduced into chain through an oxidative reaction catalyzed by FattyAcyl-CoA desaturases.
- Mammalian hepatocytes can generate unsaturations at the $\Delta^{9}$ position of fatty acids, but cannot introduce additional double bonds beyond the C10.
- Therefore, mammals cannot synthesize linoleic ($18.2\Delta^{9,12}$) or α-linolenic ($18.3\Delta^{9,12,15}$)
- On the other hand, plants can do it, so both fatty acids are essential in the human diet.
- The importance of linoleic and α-linolenic is based on the fact that they are precursors of other very important polyunsaturated fatty acids in human physiology such as arachidonic (20: $4\Delta^{5,8,11,14}$) which is an essential precursor of eicosanoids (prostaglandins, prostacyclins, thromboxanes and leukotrienes)

**Biosynthesis of Triacylglycerols (TAG)**

- The bulk of the fatty acids synthesized or ingested by humans have two possible destinations:
    - **storage** in the form of TAG, for later use as metabolic fuel
    - **incorporation** in the form of phospholipids **to the membranes**

- Whether the destination is one or the other, **depends largely on the energy needs of the body**
- During growth (childhood, adolescence) the synthesis of new membranes forces a significant fraction of the fatty acids to be used for phospholipid synthesis.
- On the other hand, when the growth phase have already passed, most of the fatty acids are incorporated as TAG into adipose tissue.
- Although most human tissues can synthesize TAG, the liver and adipose tissue have the highest synthesis rate.
- Unlike what happens with the storage of glucose as glycogen (about 400g are stored), humans can store about 15 kg of TAG
- Both TAGs and glycerophospholipids share several precursors: **glycerol 3-phosphate** (G3P) **and activated fatty acids in the form of CoA thioesters (fatty acyl-CoA)**
- Most of the G3P comes from dihydroxyacetone phosphate (DHAP) that is generated during glycolysis.
- In the liver and kidney there is also a small part of the G3P that is formed directly from glycerol through glycerol kinase
- The second precursors, the fatty acyl-CoA have their origin in the same enzymes that activate the fatty acids for their transport to the mitochondria, the acyl-CoA synthetases.
- Once the G3P is produced, the acylations (addition of fatty acids) of carbons 1 and 2 of glycerol begin. This generates phosphatidic acid (diacyl glycerol 3-phosphate) which is a precursor to both TAG and glycerophospholipids.
- In the TAG synthesis route, phosphatidic acid is dephosphorylated to DAG and finally

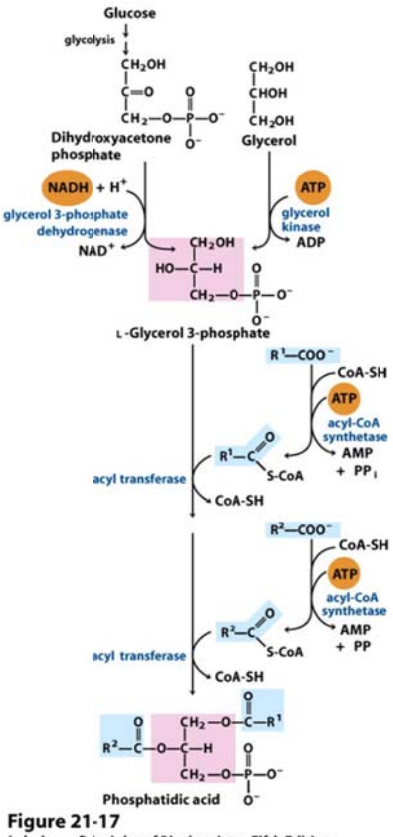



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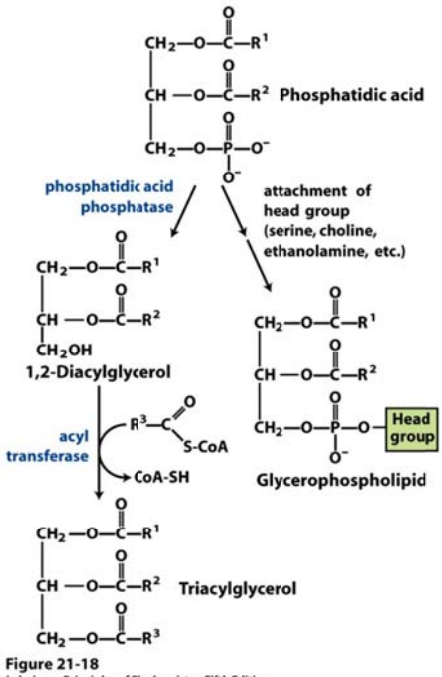
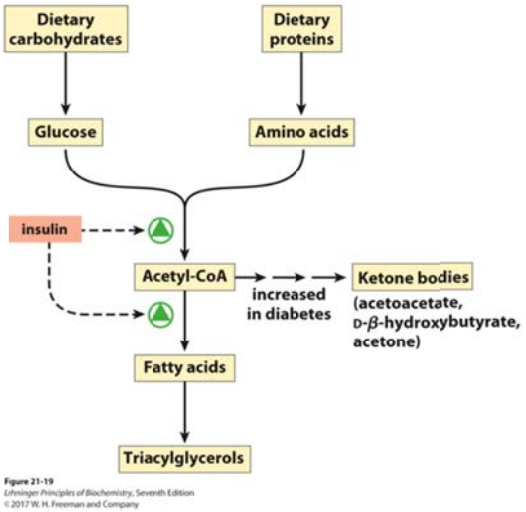


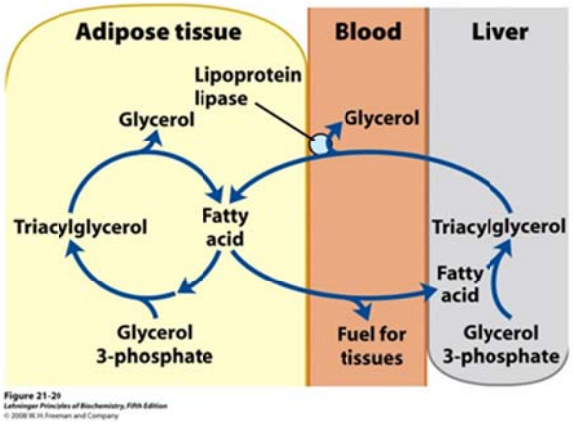

**Figure 21-18**
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C3 is acylated, giving rise to TAG

**Regulation of TAG biosynthesis**

- The biosynthesis and degradation of TAG is regulated in a way that the favored route depends on the energy needs of the moment
- Insulin, whose blood levels increase after meals (stimulated by the elevation of glucose in the blood), promotes the conversion of glucose into fatty acids and these into GAD
- Insulin-dependent diabetics, in addition to not being able to use glucose, cannot synthesize fatty acids from glucose.
- These individuals have a very high rate of fat oxidation and production of ketone bodies
- A factor that helps to balance the biosynthesis and degradation of TAG is that 75% of the fatty acids that are released during lipolysis in adipose tissue are re-esterified back into TAG.
- Part of the re-esterification occurs in adipose tissue, but another part takes place in the liver; TAGs recycled in the liver return to the bloodstream from where fatty acids (after being separated from glycerol by lipases in the endothelium) are taken up again by adipose tissue.




- Although this cycle of triacylglycerol may seem useless, the meaning may be that it allows to maintain constant levels of circulating fatty acids in the blood that can be used quickly when there are energy needs